

Multi-Member Districts and Proportional Representation: Exploring Alternatives to Single-Member Redistricting

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Abstract

1 Introduction

Congressional redistricting in the United States operates under a fundamental constraint: single-member districts elected by winner-take-all plurality voting. This combination creates structural incentives for gerrymandering while systematically excluding minor parties from representation [? ?]. The 435 single-member districts that constitute the U.S. House of Representatives ensure that even when minor parties (Libertarian, Green, etc.) receive 3-5% of the national vote, they win zero seats [?]. Meanwhile, the winner-take-all mechanism enables partisan gerrymandering through the strategic manipulation of district boundaries—packing opposition voters into supermajority districts while cracking remaining voters across districts where they form ineffective minorities [?].

This institutional design is not universal. Among established democracies, the United States is exceptional in its exclusive reliance on single-member districts for legislative elections [?]. Most parliamentary democracies employ multi-member districts (MMDs) with proportional representation (PR), allocating multiple seats per district in proportion to each party’s vote share [?]. Ireland uses 3-5 member constituencies with the single transferable vote [?]. Germany’s mixed-member proportional system combines single-member districts with party-list seats to ensure nationwide proportionality [?]. New Zealand switched from first-past-the-post to mixed-member proportional representation in 1996 following a citizen referendum, motivated by concerns about disproportional representation and wasted votes [?].

The United States has historical experience with multi-member districts. From 1789 to 1842, many states elected representatives through at-large or multi-member constituency elections [?]. The Apportionment Act of 1842 first required single-member districts, though enforcement was inconsistent. The Uniform Congressional District Act of 1967 codified the single-member requirement in federal law [?]. This institutional choice was not constitutionally mandated—the Constitution is silent on district magnitude—but rather reflects

legislative preference for geographically localized representation and incumbent protection through the two-party system [?].

This paper explores algorithmic redistricting with multi-member districts as an alternative to the current system. We design three MMD configurations for the U.S. House of Representatives using recursive bisection graph partitioning: 145 districts of 3 representatives each, 109 districts of 4 representatives, and 87 districts of 5 representatives (totaling 435 seats). For each configuration, we simulate the 2020 presidential election using party-list proportional representation with the D’Hondt seat allocation method [?]. Our analysis quantifies four dimensions: proportionality (Gallagher index), minor party representation (seat shares for parties receiving <5% national vote), minority representation (comparing proportional outcomes to single-member majority-minority districts), and compactness (Polsby-Popper ratio).

The value of this analysis extends beyond academic interest. Structural gerrymandering elimination distinguishes this approach from boundary optimization methods [? ?]. Even if MMD boundaries are drawn with partisan bias, proportional seat allocation within districts prevents packing and cracking strategies from delivering disproportional outcomes. The system becomes effectively ungerrymandered not through careful boundary design but through the proportional allocation mechanism itself. Additionally, proportional representation addresses Voting Rights Act compliance in a fundamentally different way than single-member majority-minority districts [?]. Rather than concentrating minority populations geographically to create majority-minority districts, proportional allocation enables minority representation through vote share: a district with 35% Black voters electing 5 representatives would typically elect 2 Black representatives under cohesive voting patterns, without requiring geographic concentration that critics argue constitutes "packing" [?].

Our contributions are threefold. First, we provide the first algorithmic design of multi-member districts for the U.S. House using recursive bisection, extending prior work on single-member algorithmic redistricting [? ?] to the MMD context. Second, we quantify the proportionality-compactness trade-off: larger multi-member districts (5 representatives) achieve near-perfect proportionality (Gallagher index < 2.0) but at the cost of reduced compactness (approximately 10% loss in Polsby-Popper ratio). Third, we demonstrate that 5-member MMDs enable minor party representation—parties receiving 3-5% of the vote within a district can win seats—while 3-member MMDs remain effectively two-party systems due to high exclusion thresholds [?].

- 2 Related Work
- 3 Methodology
- 4 Evaluation
- 5 Discussion
- 6 Conclusion